

	$= 1210 \text{ m/s}$
3(a)	A polarised wave is a transverse wave in which its oscillations are only in one direction perpendicular to the direction of propagation of the wave.
3(b)	For a progressive wave that travels with a constant speed, the distance travelled by the wave in one period T is one wavelength λ. Since speed is defined as the distance travelled per unit time, the speed v of the wave is $v = \frac{\lambda}{T} = f\lambda$
3(c)(i)	Using Pythagoras' Theorem, the distance BP is 13 km. The path difference between AP and BP is 1 km. $\lambda = \frac{3.0 \times 10^8}{1200 \times 10^3} = 250 \text{ m}$ Hence, the path difference is equivalent to 4 wavelengths. Since the transmitters are in phase, and the path difference is an integer number of complete wavelengths, the waves will meet in phase at P. This will result in constructive interference.
3(c)(ii)	The resultant amplitude will vary between maxima and minima with a constant frequency as the car moves at a constant speed.
3(c)(iii)	The waves from the transmitters can be polarised in perpendicular directions so that they are not able to produce an interference pattern.

3(d)	$I_A = \frac{P}{4\pi (12)^2}$ $I_B = \frac{P}{4\pi (13)^2}$ $\frac{I_A}{I_B} = \left(\frac{13}{12}\right)^2 = 1.17$
4(c)	